

NATURAL LANGUAGE PROCESSING

Foundations, Web Scraping, Sentiment Analysis & Applications

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Computer Engineering Program
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Universidad Distrital Francisco José de Caldas

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UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Foundations and Basic Concepts
- 2 Web Scraping and Data Collection
- 3 Sentiment Analysis
- 4 Applications and Integration



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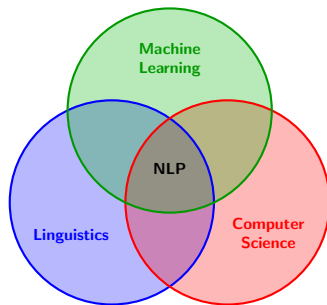
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What is Natural Language Processing?

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Why NLP is Hard:

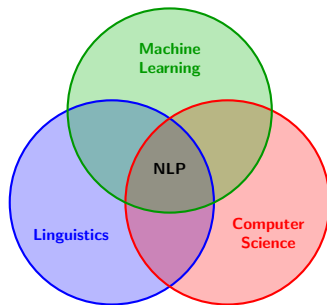
- *Ambiguity: "I saw the man with the telescope" — who has it?*
- *Context: meaning depends on co-text and world knowledge*
- *Learning: learning from examples*



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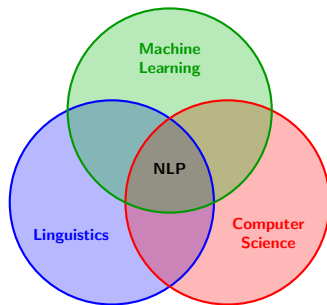
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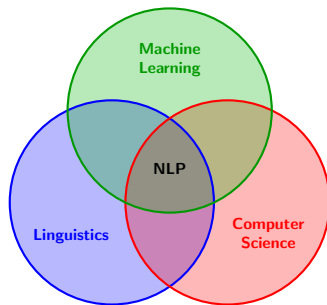
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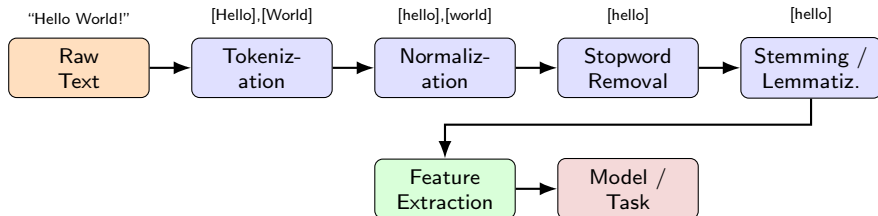


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The NLP Processing Pipeline



- **Tokenization:** split into **words**, subwords (BPE), or characters
- **Normalization:** lowercase, accent removal, punctuation stripping
- **Stemming:** *running*→*run*; **Lemmatization:** *better*→*good* (grammatically correct root)



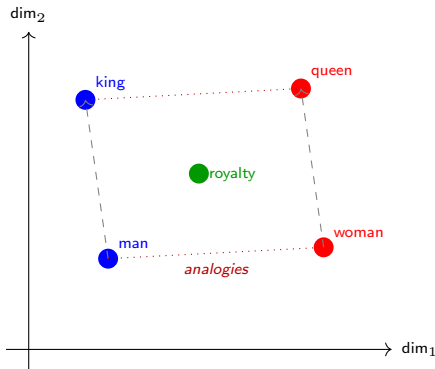
Text Representation: From Words to Vectors

Bag-of-Words & TF-IDF

- Doc $\rightarrow \mathbf{v} \in \mathbb{R}^{|V|}$; **order ignored**
- $\text{tfidf}(t, d) = \text{tf} \cdot \log \frac{|D|}{df_t}$
- Downweights **common**, upweights **rare** terms

Word Embeddings

- Dense $\mathbb{R}^d$; semantic similarity $\approx$ distance
- $\text{king} - \text{man} + \text{woman} \approx \text{queen}$



Word2Vec 2-D projection



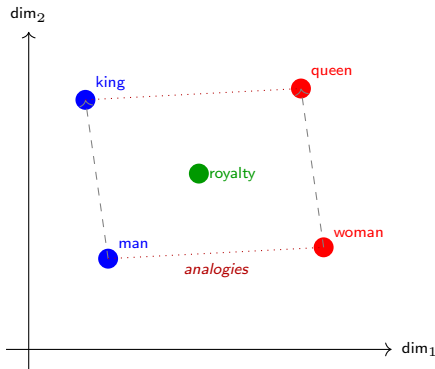
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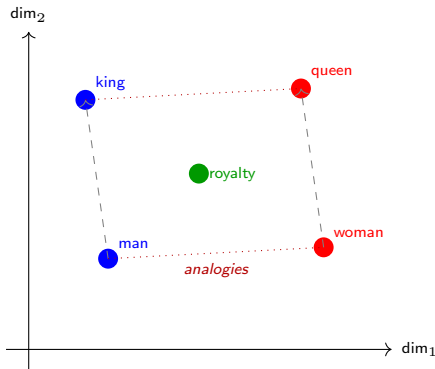
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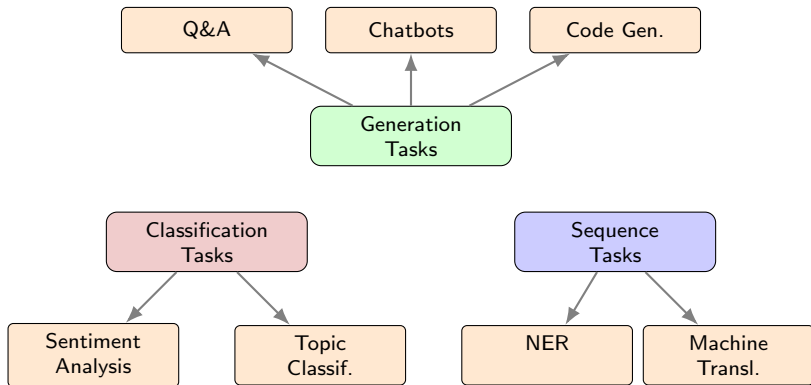
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Key NLP Tasks Overview



BERT, GPT, T5: one model, all tasks via fine-tuning



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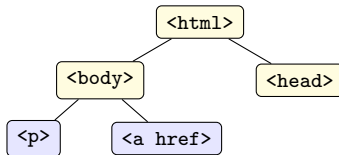
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The HTTP Request — Response Cycle



Parsed HTML Tree



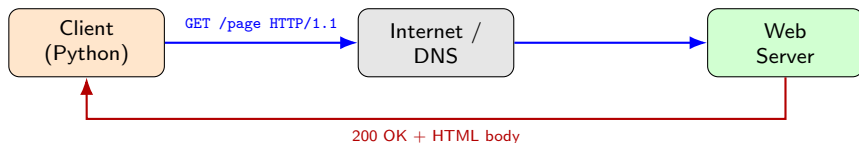
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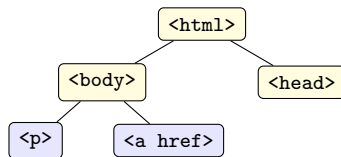
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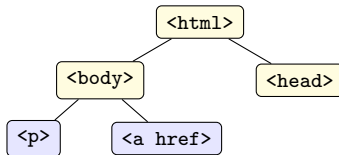
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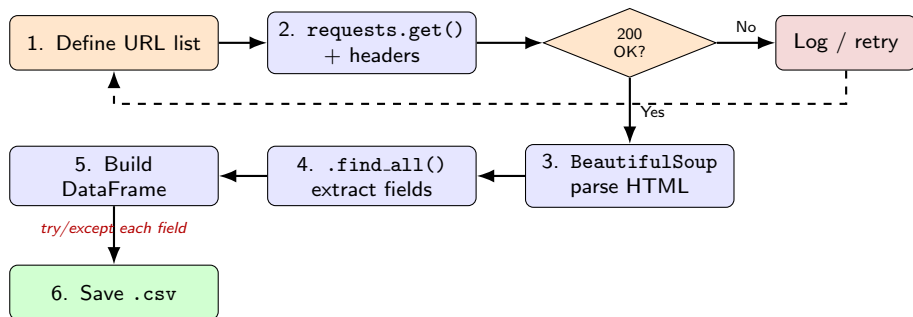
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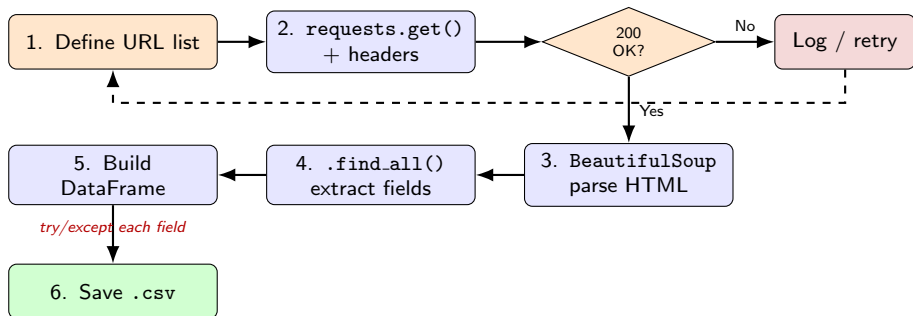
Scraping Workflow with requests + BeautifulSoup



- Add `time.sleep(1--3)` between requests (**politeness** delay)
- Always respect `robots.txt` and Terms of Service



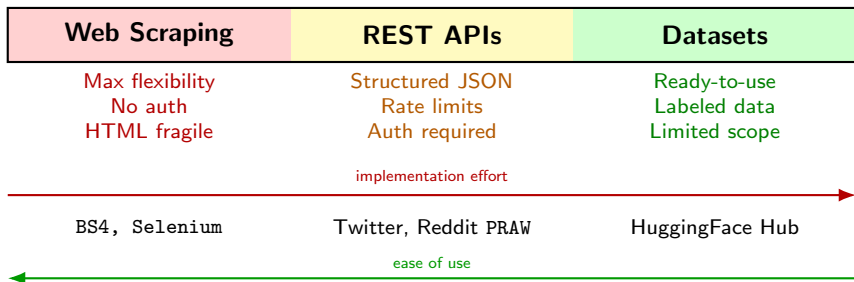
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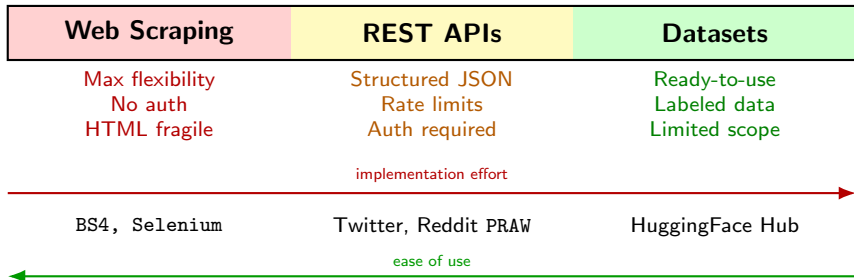


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Check `robots.txt`, Terms of Service, and GDPR constraints. Do **not** collect personal data without consent.



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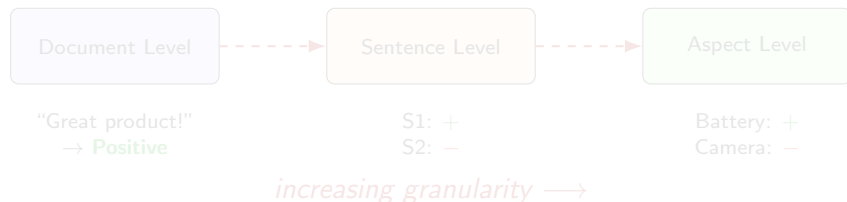
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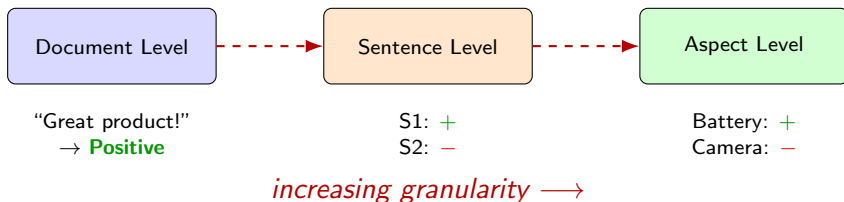
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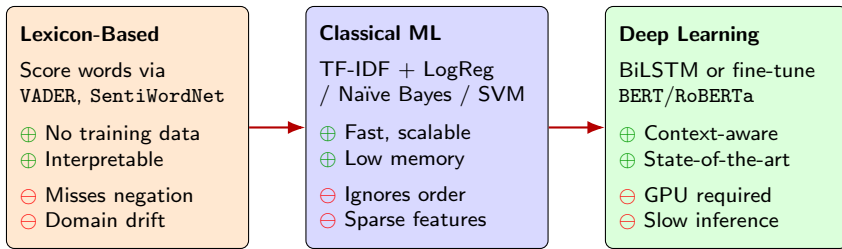
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Three Approaches: Lexicon, ML, Deep Learning



accuracy & data requirements increase →



Evaluation: *Metrics and Confusion Matrix*

Beyond Accuracy

Accuracy is misleading on **imbalanced** data. Always report per-class metrics.

- Precision = $\frac{TP}{TP + FP}$
- Recall = $\frac{TP}{TP + FN}$
- F1 = $\frac{2 \cdot P \cdot R}{P + R}$
- F1-macro: average over all classes

Confusion Matrix

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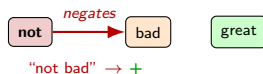
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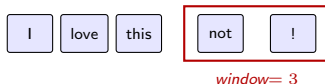
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- **Negation:** “*Not bad*” → **positive**
- **Sarcasm:** “*Great, another Monday*”
- **Domain shift:** medical $\neq$ movie vocabulary
- **Multilingual:** cultural conventions differ
- **Subjectivity:** “*Film runs 120 min*” has **no sentiment**

Negation Scope



Context Window



Practical Tip

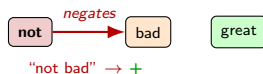
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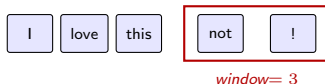
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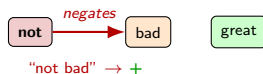
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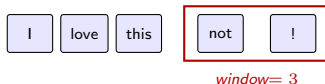
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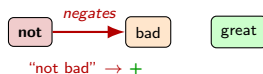
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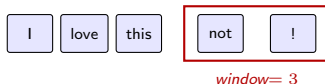
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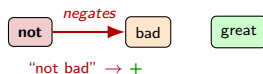
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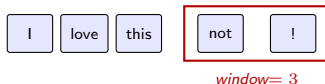
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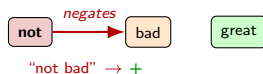
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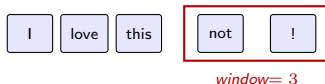
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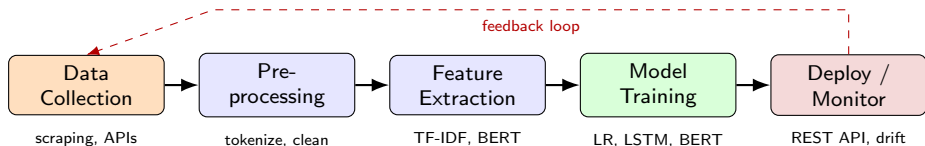


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End-to-End NLP Pipeline

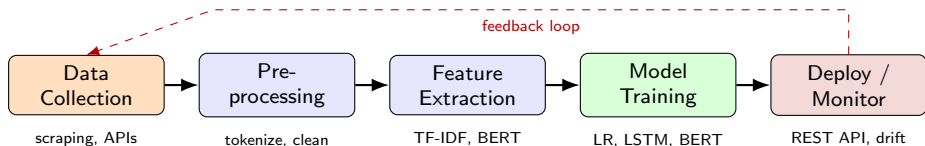


Real-World Applications:

- Chatbots & virtual assistants
- Machine translation (DeepL)
- Search engine ranking
- NER in clinical records
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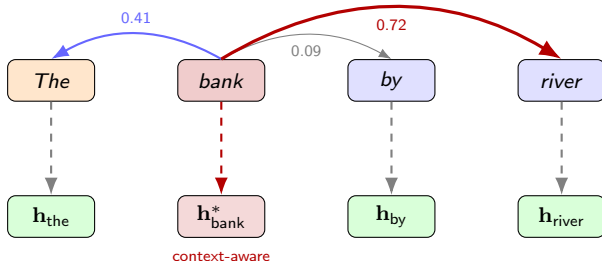


Transformer Models: Self-Attention

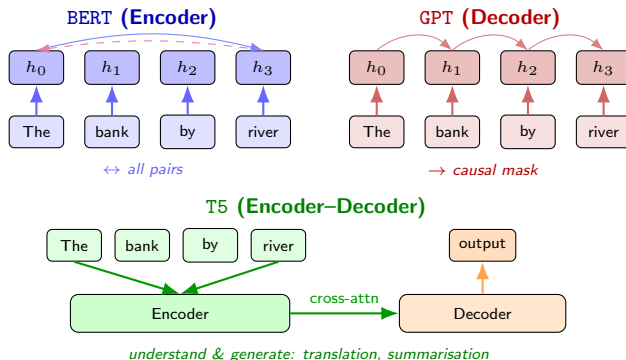
Key Innovation (Vaswani *et al.*, 2017)

Replace recurrence with **attention**: every token simultaneously queries **all** other tokens, capturing **long-range context** in parallel.

“bank” attends most to “river” → river bank, not financial bank



Transformer Families: Architecture Comparison



BERT pre-training

MLM: predict masked tokens

"The [MASK] by the river"

GPT pre-training

CLM: predict next token

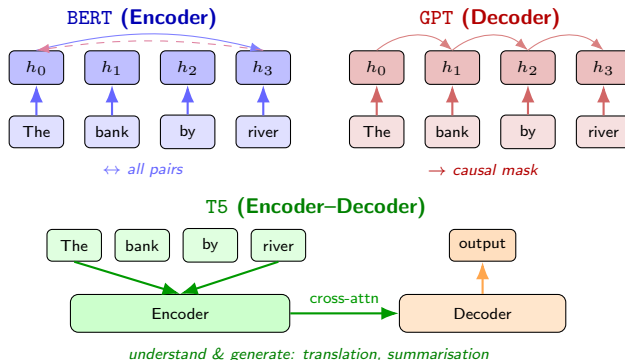
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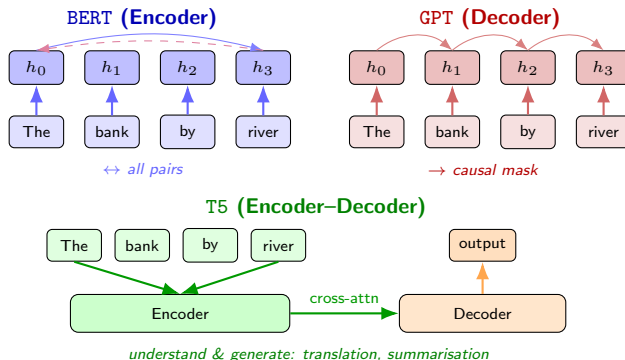
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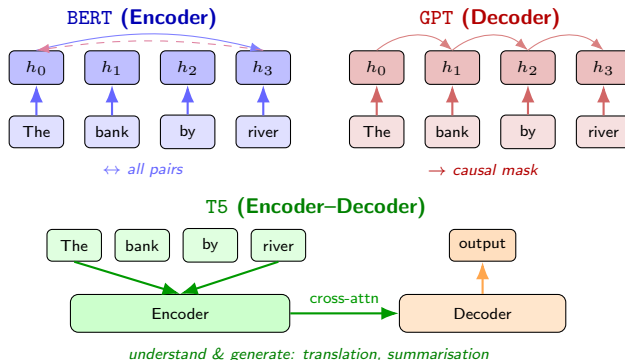
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Encoding a Sentence

- Feed full sentence through BERT
- Pool [CLS] token or mean-pool hidden states
- Output: $\mathbf{v} \in \mathbb{R}^{768}$ — a semantic fingerprint

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$$\cos(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

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Semantic search in embedding space

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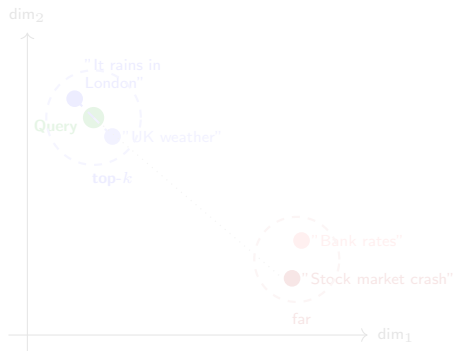
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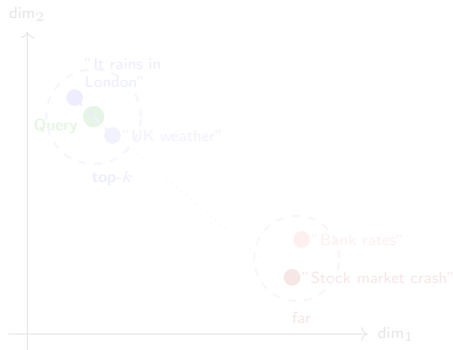
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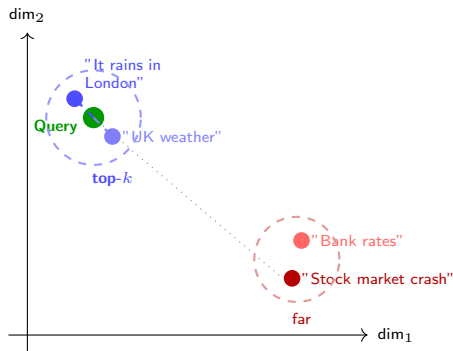
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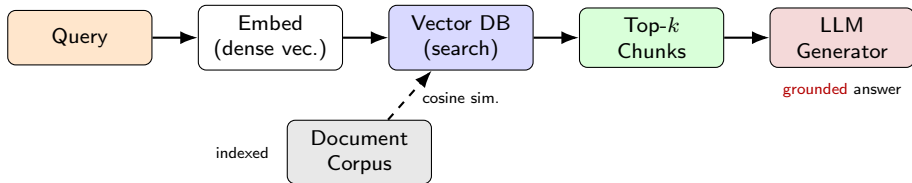
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Retrieval-Augmented Generation (RAG)

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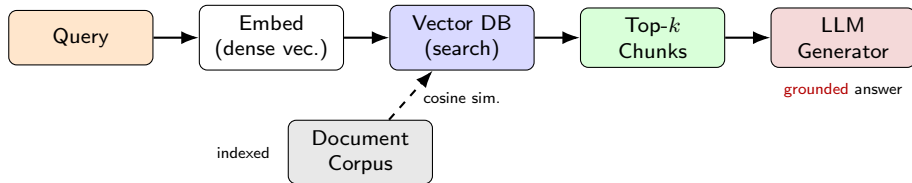
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Current Challenges in NLP

- **Hallucination:** LLMs generate **plausible but false** content
- **Bias:** stereotypes inherited from **training corpora**
- **Low-resource languages:** $\sim 7,000$ languages; most resources are **English-centric**
- **Interpretability:** SHAP, LIME, Attention maps as post-hoc explanations
- **Computational cost:** GPT-4 $\approx$ \$200M to train; LoRA/PEFT for efficient fine-tuning
- **Privacy:** text often contains PII — **GDPR** compliance required

Workshop Connection

Build a complete pipeline: scrape *reviews* $\rightarrow$ *preprocess* $\rightarrow$ *train sentiment classifier* $\rightarrow$ *deploy* as a Streamlit app with live **inference**.



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Outline

- 1 Foundations and Basic Concepts
- 2 Web Scraping and Data Collection
- 3 Sentiment Analysis
- 4 Applications and Integration



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/data-analysis-programming>

